Matrix Decompositions

Math 5250 — Week 7

Overview

Consequences of the spectral theorem and of our study of unitary and orthogonal matrices that are important in applications:

- the LU decomposition (Gaussian elimination, for reference);
- the QR decomposition, with applications to least squares and to computing eigenvalues;
- the polar decomposition $L = UP$.

The LU Decomposition

The LU Decomposition

Proposition. Let A be an $n \times m$ matrix over any field. Then there exist

- an $n \times n$ permutation matrix P,
- an $n \times n$ lower triangular matrix L,
- an $n \times m$ upper triangular matrix U ($u_{ij} = 0$ when $i > j$),

such that

$$PA = LU.$$

This has nothing to do with the spectral theorem: it is a restatement of Gaussian elimination.

Proof: Gaussian Elimination

Apply row operations to A until it is upper triangular. Generically each diagonal entry in turn serves as a pivot, and each row operation is left multiplication by an elementary matrix E_i that is lower triangular with ones on the diagonal:

$$E_k E_{k-1} \cdots E_1 A = U, \qquad L = (E_k E_{k-1} \cdots E_1)^{-1}.$$

- In general, rows may need to be permuted to get a nonzero pivot; collect these into P and replace A by PA.
- If A is singular (or not square), U may have zeros on its diagonal.

The QR Decomposition

The QR Decomposition

Let X be an $n \times k$ matrix with real or complex entries and linearly independent columns.

Proposition. There is an $n \times n$ orthogonal (or unitary) matrix Q and an $n \times k$ upper triangular matrix R such that

$$X = QR.$$

This is just a version of the Gram–Schmidt algorithm.

QR from Gram–Schmidt

Gram–Schmidt applied to the columns $x_1, \ldots, x_k$ of X gives orthonormal $u_1, \ldots, u_k$ with $u_j \in \operatorname{span}(x_1, \ldots, x_j)$. With $U = [u_1 \ \cdots \ u_k]$,

$$U = XS, \qquad S \ k \times k \text{ upper triangular},$$

so $X = US^{-1}$.

- Extend U by orthonormal columns to an orthogonal/unitary $n \times n$ matrix Q.
- Add $n - k$ zero rows to S^{-1} to get an $n \times k$ upper triangular R.

Then $X = QR$.

Application: Least Squares

Given an $N \times k$ data matrix X (rows are samples, columns are features) and targets $Y \in \mathbb{R}^N$, we model

$$y_i = \sum_{j=1}^{k} m_j x_{ij} + b + \epsilon_i.$$

Let $M = (m_1, \ldots, m_k, b)^\top$ and let X' be X with a column of 1's appended. Then

$$Y = X'M + \epsilon.$$

The M minimizing $S(M) = \|Y - X'M\|^2$ solves the **normal equations**

$$(X')^\top Y = (X')^\top X' M.$$

Least Squares via QR

Use Gram–Schmidt to write $X' = UK$, where U has orthonormal columns and K is $(k+1) \times (k+1)$ upper triangular. Then

$$(X')^\top X' = K^\top U^\top UK = K^\top K,$$

since $U^\top U = I$.

If the columns of X' are independent, K is invertible, and the normal equations become

$$U^\top Y = KM.$$

Since K is upper triangular, solve by **back substitution**.

The QR Algorithm

Motivation: Power Iteration

Let A be a positive real symmetric $n \times n$ matrix with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and orthonormal eigenvectors $u_1, \ldots, u_n$.

Choose a random unit vector v_0 and iterate

$$v_{k+1} = \frac{Av_k}{\|Av_k\|}.$$

Then v_k converges (up to sign) to u_1, the eigenvector for the largest eigenvalue λ_1.

Why Power Iteration Works

Write $v_0 = \sum_i a_i u_i$ with $a_1 = \langle v_0, u_1 \rangle \neq 0$. Then v_k is the unit vector in the direction of

$$\begin{aligned} A^k v_0 &= \sum_{i=1}^{n} a_i \lambda_i^k u_i \\ &= \lambda_1^k \sum_{i=1}^{n} a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k u_i = \lambda_1^k (a_1 u_1 + e_k), \end{aligned}$$

where $e_k \to 0$ at least as fast as $(\lambda_2/\lambda_1)^k$.

So v_k aligns with u_1 as $k \to \infty$.

The QR Algorithm

Same idea, but apply Gram–Schmidt to *all* columns of A^k, so each column converges to an eigenvector.

Algorithm. Let $A_0 = A$ be $n \times n$ real symmetric. For $k \geq 0$:

1. Compute the QR decomposition $A_k = Q_k R_k$.
2. Set $A_{k+1} = R_k Q_k$.

Since $A_{k+1} = Q_k^{-1} A_k Q_k$, all the A_k are orthogonally similar to A and have the same eigenvalues.

Accumulated Factors

Put

$$\overline{Q}_k = Q_0 Q_1 \cdots Q_{k-1}, \qquad \overline{R}_k = R_{k-1} R_{k-2} \cdots R_0.$$

Then

$$A_k = \overline{Q}_k^\top A \, \overline{Q}_k \qquad \text{and} \qquad A^k = \overline{Q}_k \overline{R}_k.$$

- $\overline{Q}_k$ is Gram–Schmidt applied to the columns of A^k.
- A_k converges to a diagonal matrix of eigenvalues of A.
- $\overline{Q}_k$ converges to a matrix whose columns are eigenvectors of A.

Example

$$A = \begin{pmatrix} 2.1948 & 0.5638 \\ 0.5638 & 2.6052 \end{pmatrix}$$

- The level curves $u^\top A_k u = 1$, $u = (x, y)^\top$, are ellipses, each a rotation of the previous one.
- As A_k becomes diagonal, the axes of the ellipse line up with the x- and y-axes.
- The off-diagonal entry of A_k decays like $(\lambda_2/\lambda_1)^k$: its logarithm decreases linearly with slope $\log(\lambda_2/\lambda_1)$.

(See the animation in the online notes.)

Convergence

Proposition. Let A be a positive definite real symmetric $n \times n$ matrix with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$. Then $A_k \to \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, with rate $O(\rho^k)$, where

$$\rho = \max_j \frac{\lambda_{j+1}}{\lambda_j}.$$

We sketch why, assuming (generically) that V^{-1} below has an LU decomposition with no permutation.

Sketch of Proof (1)

Write $A = V\Lambda V^\top = V\Lambda V^{-1}$ with V orthogonal, and $V^{-1} = LU$. Then

$$A^k = V\Lambda^k LU = V\left(\Lambda^k L\Lambda^{-k}\right)\Lambda^k U.$$

The (i,j) entry of $\Lambda^k L\Lambda^{-k}$ is $\ell_{ij}(\lambda_i/\lambda_j)^k$. For $i > j$ this ratio is < 1, so

$$\Lambda^k L\Lambda^{-k} = I + E_k, \qquad E_k = O(\rho^k).$$

Sketch of Proof (2)

Take the QR decomposition $I + E_k = \tilde{Q}_k \tilde{R}_k$. Since Gram–Schmidt is continuous and $I = I \cdot I$,

$$\tilde{Q}_k \to I, \qquad \tilde{R}_k \to I.$$

Then

$$A^k = \underbrace{V \tilde{Q}_k}_{\text{orthogonal}} \quad \underbrace{\tilde{R}_k \Lambda^k U}_{\text{upper triangular}}$$

is a QR decomposition of A^k.

Sketch of Proof (3)

The QR decomposition is unique up to the signs of the columns, so

$$\overline{Q}_k = V \tilde{Q}_k D_k, \qquad D_k = \text{diag}(\pm 1).$$

Therefore

$$\begin{aligned} A_k = \overline{Q}_k^\top A \overline{Q}_k &= D_k \tilde{Q}_k^\top (V^\top A V) \tilde{Q}_k D_k \\ &= D_k \tilde{Q}_k^\top \Lambda \tilde{Q}_k D_k. \end{aligned}$$

Since $\tilde{Q}_k \to I$ and $D_k \Lambda D_k = \Lambda$, we get $A_k \to \Lambda$. $\square$

The Polar Decomposition

Polar Form

For a complex number $z \neq 0$,

$$z = re^{i\theta}, \qquad r > 0,$$

with θ determined up to a multiple of 2π.

The polar decomposition generalizes this to linear operators:

$$L = UP, \qquad P \text{ positive}, \quad U \text{ unitary}.$$

Square Roots of Positive Operators

Recall that a self-adjoint L is **positive** if $\langle Lv, v\rangle \geq 0$ for all v.

Such an L is unitarily (orthogonally) diagonalizable with nonnegative eigenvalues $\lambda_1, \ldots, \lambda_k$ and spectral resolution

$$L = \sum_{i=1}^{k} \lambda_i p_i,$$

where p_i is the orthogonal projection onto the λ_i-eigenspace.

Definition. The square root of L is

$$\sqrt{L} = \sum_{i=1}^{k} \sqrt{\lambda_i}\, p_i \qquad (\sqrt{\lambda_i} \geq 0).$$

Then $(\sqrt{L})^2 = L$, and $\sqrt{L}$ is the unique positive operator whose square is L.

Positive Operators are A^*A

For any linear $A : V \to V$, A^*A is self-adjoint and positive:

$$\langle A^*Av, v\rangle = \langle Av, Av\rangle = \|Av\|^2 \geq 0.$$

Conversely, if L is positive and $A = \sqrt{L}$, then $L = A^*A$.

Proposition. L is positive if and only if $L = A^*A$ for some operator A.

Example. The matrix $X^\top X$ from least squares regression is positive.

The Polar Decomposition

Proposition. Let $L : V \to V$ be a linear operator on a finite-dimensional real or complex inner product space. Then there exist a positive P and a unitary (orthogonal) U with

$$L = UP.$$

P is unique, and if L is invertible, U is unique.

Similarly $L = P'U'$ with P' positive and U' unitary (orthogonal); P' is unique, and U' is unique if L is invertible.

Constructing P

If $L = UP$, then

$$L^*L = P^*U^*UP = P^2,$$

so by uniqueness of positive square roots,

$$P = \sqrt{L^*L}.$$

Constructing U on $\mathrm{image}(P)$

If $w = Pv \in \mathrm{image}(P)$, define $U(w) = L(v)$.

Well-defined:

$$\|Pv\|^2 = \langle P^2 v, v\rangle = \langle L^* L v, v\rangle = \|Lv\|^2,$$

so $Pv_1 = Pv_2$ implies $L(v_1 - v_2) = 0$.

Linear: $U(aPv + Pv') = aLv + Lv' = aU(Pv) + U(Pv')$.

Isometry: $\|Uw\| = \|Lv\| = \|Pv\| = \|w\|$.

So U is an isometry from $\mathrm{image}(P)$ onto $\mathrm{image}(L)$, and these have the same dimension h.

Extending U to V

- Choose an orthonormal basis $u_1, \ldots, u_h$ of $\mathrm{image}(P)$; then $U(u_1), \ldots, U(u_h)$ is an orthonormal basis of $\mathrm{image}(L)$.
- Extend to an orthonormal basis $u_1, \ldots, u_n$ of V.
- Choose an orthonormal basis $w_{h+1}, \ldots, w_n$ of $\mathrm{image}(L)^{\perp}$.
- Set $U(u_j) = w_j$ for $j > h$.

U carries an orthonormal basis to an orthonormal basis, so it is unitary (orthogonal), and $L = UP$.

If L is invertible, so is P, and $U = LP^{-1}$ is determined.

Unitary Operators are Exponentials

Proposition. Over $\mathbb{C}$, every unitary U has the form $U = \exp(iJ)$ with J self-adjoint.

Proof. U is unitarily diagonalizable with eigenvalues of absolute value 1, i.e. of the form $e^{i\theta}$ with θ real. Let J act by θ on the eigenspace where U acts by $e^{i\theta}$. Then $U = \exp(iJ)$, and J is unitarily diagonalizable with real eigenvalues, hence self-adjoint. $\square$

So $L = UP = e^{iJ}P$ is the operator analogue of $z = e^{i\theta}r$.